

RQ: To what extent did changes in Federal Reserve interest rates influence JP Morgan's financial performance during 2018-2024?

1.1) Introduction

The US Federal Reserve interest rate plays a pivotal role in influencing financial decisions across the USA, affecting borrowing rates, liquidity, investment behaviour, and profitability. Large diversified banks like JP Morgan Chase & Co tend to benefit during a tightening cycle :variable rate loans and short-term securities reprice swiftly in comparison to deposits. A large portion of all its deposits are non-interest bearing and are unresponsive towards shifts in interest rates (Drechsler et al., 2017).

Between 2018 and 2019 the Fed implemented a modest tightening cycle with interest rates averaging around 2% between 2018 & 2019. The Fed changed course and implemented expansionary policies in the pandemic that brought the interest rate to a floor of 0.05% in 2020 (FRED). After the pandemic the interest rate rose quickly and reached 5.33%.

JP Morgan is selected for three reasons. Firstly, it is the largest bank in the USA by assets. Secondly, it is designated as a G-SIB (Globally Systemically Important Bank), meaning that its financial performance has implications beyond the firm's balance sheet.

Lastly, JP Morgan's resilience reflects revenue diversification: ~ half of its revenue comes from markets, investment banking fees and asset and wealth management. These firms respond to high volatility and deal flows rather than the policy rate and can affect the cyclicity of the firm's Net Interest Income (NII)

Hence this paper investigates the research question "To what extent did changes in Federal Reserve interest rates influence JP Morgan's financial performance during 2018-2024?".

1.2) Significance

This study makes three significant contributions to the existing literature.

The theoretical link between interest rate variations and bank profitability is established by (Borio et al., 2017; English et al., 2018). However, these studies focus primarily on cross-sectional samples or on historical periods where there have been gradual rate movements. In contrast, this paper uses three distinct time frames with different monetary policies. This allows for extensive analysis of how interest rates affect bank profitability.

Empirically, the single firm focus on a G-SIB provides quantitative depth that cross-sectional studies cannot offer. Moreover, evaluating its asset-liability composition, deposit sensitivity, loan repricing behaviour highlights how internal mechanisms influence its financial performance.

In application, this analysis has significance for two main audiences. For policymakers it illustrates how large banks absorb monetary shocks. This helps in anticipating the consequences of tightening and loosening cycles. On the other hand, for banks it provides a framework to optimize their balance sheets during both monetary tightening and easing periods, with the aim of preserving profitability and solvency. On a broader note due to the fact that JP Morgan is the largest G-SIB, its experience provides evidence as to why such a bank is insulated from rate shocks rather than as a template that banks can use. Thus this study's natural beneficiaries are analysts, bank risk and ALM teams.

1.3) Methodology

Variables:

Independent Variable (IV): Federal Reserve Interest Rates

Dependent Variable (DV): Five Indicators of Financial Performance: (i) Pre-Provision Net Revenue relative to assets (earnings), (ii) the CET1 ratio (capital), (iii) the Net charge-off rate (asset quality), (iv) the Liquidity Coverage Ratio (liquidity), and (v) Unrealised Securities Losses-to-CET1 (Available-for-sale (AFS) losses — each scaled by CET1) (Securities).

These 5 DVs are synthesized into a single Bank Financial Performance Index (BFPI) described under Analytical Methods.

Analytical methods: The primary approach utilized is quantitative comparative analysis around the three monetary cycles: pre pandemic (2018-2019); zero lower bound period (2020-2021) ;post pandemic (2022-2024). For each time frame JP Morgan's financial performance is measured and evaluated with relation to the interest rates. This time-frame analysis provides a detailed understanding of the firm's balance sheet across distinct monetary regimes.

To synthesize the dependent variables into a single comparable index, this study constructs an original indicator—Bank Financial Performance Index (BFPI). The BFPI is custom-tailored for this paper and does not draw from any existing literature. Moreover, this index is benchmarked against Return on Tangible Common Equity (ROTCE)

Each Indicator is standardized against the JP Morgan's Historical Data using a robust z score that is built from the median and the median absolute deviation (MAD).

where:

- $X_{K,T}$ =JP Morgan reported value of indicator K in quarter T
- M_k = Median of JP Morgan's Quarterly Values of indicator k across 2018-2024
- MAD_k = Median Absolute Deviation of indicator K
- 1.4826 is the scaling constant that rescales the MAD into a consistent estimator of the standard deviation under normality
- D_k = A direction coefficient that sets a direction for the sign of each indicator. $DK = +1$ indicators where positive value denote stronger performance (earnings, capital, liquidity) and $DK = -1$ for whose higher values denote weaker performance (asset-quality deterioration and rising securities losses).

The median is utilized as a measure of central tendency for two main reasons. Firstly, it is the 0.5 quantile and it is the minimiser of the sum of absolute deviation, making the location estimator under an absolute-error criterion. Secondly, it attains the maximal breakdown point of 50%— a theoretical ceiling for any location estimator.

Whereas the Median Absolute Deviation is utilized as a measure of dispersion for 2 main reasons. Primarily, as a quantile of the absolute deviation distribution, it has a maximal 50% breakdown point and also has a bounded influence function, so no outliers can affect the data.

Aggregation: The index is equal-weighted mean of the five standardised indicators:

Where:

- BFPI - Composite Index Value for Quarter t .
- Z^* - standardised score of each indicator from the standardization equation
- Σ (from $k = 1$ to 5) – adds the five standardized score from quarter t — multiplying its sum by $\frac{1}{5}$ provides the average— thereby, the BFPI is the mean of the five standardised scores.

In order to maintain robustness the BFPI has assigned an equal weighting to each component to ensure that macroeconomic conditions do not affect the assignment of weights.

Interpretation – The BFPI is expressed in standard-deviation units of JP Morgan's composite performance. Moreover, each indicator has a sample median of zero, to ensure that the index has a benchmark of 0 for JP Morgan's operating state. Thereby, a positive BFPI indicates composite performance above the benchmark; conversely, a negative BFPI indicates a performance below the benchmark.

Additionally, as a supporting statistical analysis Spearman's rank correlation is used to quantify a relationship between the dependent and independent variable—strengthening the evidential basis of the relationship between interest rates and financial performance. Moreover, Spearman's

correlation is preferred over Pearson due to the relationship between rates and bank performance is plausibly non-linear and monotonic. Therefore, Spearman's rank is utilized as a tool to quantify a relationship between the variables.

Data Sources: The primary used source will be the JP Morgan Quarterly Investor report. These reports detail the company's financials for the quarter alongside the financials, an extensive evaluation of the business operations, assets, risk exposure and liquidity. Besides the investor reports, the main source is the Federal Reserve Economic Data(FRED) to obtain federal funds rate, yield curves, inflation, and unemployment.

Limitations: This study has three main limitations. Firstly the sole effect of interest-rates on financial performance cannot be isolated from macroeconomic conditions. The Fed adjusts interest rate as a response to inflation and unemployment which can independently affect loan demand, the asset quality and credit risk. Therefore, the relationship between the variables is associational and not causal. Secondly the analysis has limited 28 observations which can reduce the reliability of the correlation. Moreover, the reported p-values computed with independence assumption can overstate the statistical significance. Finally, the BFPI has element of methodological subjectivity as the choice to standardize against the median create a retrospective index and the equal weighting— although chosen for robustness is an assumption rather than a mathematically derived result.

Section 2 – Theoretical Framework

2.1) Definition of Financial Performance

Financial performance in the context of a commercial banking institution refers to a firm's ability to sustainably generate returns from its principal operations—while maintaining an adequate amount of buffer capital and efficiently managing credit risk. Moreover, for financial firms the spread between asset yields and funding is the primary determinant of profitability.

This paper evaluates JP Morgan's financial performance across five indicators that are synthesized into the BFPI.

Pre-Provision Net Revenue Relative to Assets (PPNR) - Earnings (E)

Pre-Provision Net Revenue is the income that a bank earns from its principal operations before deducting allocations for credit losses.

CET1 Ratio - Capital (C)

Common Equity Tier 1 Capital is the highest quality of regulatory capital, mainly equity and retained earnings. This is expressed as a percentage of a bank's capital relative to its risk weighted asset. It is a measure of how much the capital buffer can absorb losses before the firm reaches insolvency. It is also an essential binding measure for solvency under Basel III requirements.

Net Charge-off Rate - Asset Quality (A)

Net Charge off rate (NCO) is a metric used to measure the percentage of loans that are uncollectible and are written off as a loss.

Liquidity Coverage Ratio - Liquidity (L)

The liquidity coverage ratio is a regulatory mandate that all banks must maintain a sufficient buffer of high quality liquid assets in order to survive an intense 30 day financial stress scenario.

Unrealised Securities Losses-to-CET1(AFS/CET1) - Unrealised Securities Losses (S)-

This ratio measures the size of banks capital to its unrealised losses on its securities portfolio relative CET1. This indicates the extent to which the decline in the market value of securities erodes the quality of banks capital if the losses are realised. HTM Latent loss are used as separate exposure in 2.2.3.

2.2) Interest Rate Transmission Mechanism

The Federal Reserve set a benchmark interest rate range as its main monetary policy instrument. However these decisions are not arbitrary, rather they are guided by the Taylor Rule— a monetary policy framework that provides a normative benchmark for where the interest rates should be set in response to inflation and economic output (Taylor, 1993). The rule is utilized

here to establish that the rate environment across the time frames was a rule base yet constantly rate-deviation: the actual policy rates were well below the prescribed benchmark across all three regimes (Table 1), reflecting the Fed’s discretion in deviation from the normative benchmark due to the macroeconomic conditions.

$$i^* = r^* + \pi + a(\pi - \pi^*) + b(y - y^*)$$

i^* = Prescribed Interest Rate

r^* = Neutral Real Rate

π = Actual Inflation (CPI,%)

π^* = Inflation Target

$(y - y^*)$ = Output Gap (% of potential GDP)

$a = b = 0.5$ (Taylor’s original weights)

Table 1: Taylor Rule Prescribed Rate by Monetary Regime (2018–2024)

Period	r^* (Neutral Real Rate)	(Average Inflation (CPI,%))	π^* (Inflation Target)	$\pi - \pi^*$ (%)	$(y - y^*)$ Output Gap	α (Taylor’s weights)	β (Taylor’s weights)	i^* (Prescribed Interest Rate)
2018-2019	2	2.13	2	0.13	0.30	0.5	0.5	4.35
2020-2021	2	2.97	2	0.97	-1.50	0.5	0.5	4.71
2022-2024	2	5.02	2	3.02	2.40	0.5	0.5	9.71

Table 1 applies the Taylor rule to FRED’s macroeconomic data across the 3 monetary regimes, providing a prescribed rate for each time frame.

During the pre-pandemic tightening cycle (2018-2019) the Taylor rule suggested a 4.35% interest rate against the actual interest rate of around 2%—a 2.35 point deviation, the smallest across all three regimes. Apart from that the inflation was at around 2.13% and there was a positive output gap of 0.30. This indicates that the economy was operating slightly above its potential output. Assets yields repriced upwards and the deposits on non-interest bearing assets increase, while bearing deposits stay near zero, widening funding spread.

The zero lower bound period (2020-2021 regime) provides the most analytically rich regime. With the average CPI at 2.97% and a negative output gap indicating high levels of unemployment and low economic growth in the economy. The Taylor rule prescribed an interest rate of 4.71%, whereas the actual interest rates reached an all-time low of around 0.05%, a policy gap of 4.66 points, the largest deviation in the study. Moreover, the negative output gap provides rationale for the extreme easing cycle —stimulating consumption and reducing spare capacity to drive business and consumer confidence. However, this directly reduces the PPNR a component of BFPI through the deposit rate floor mechanism pinpointed by Claessens, Coleman, and Donnelly (2018)— with assets yield on new loans repriced to near zero, also since the non interest deposits are already at zero cost and cannot potentially fall any further.

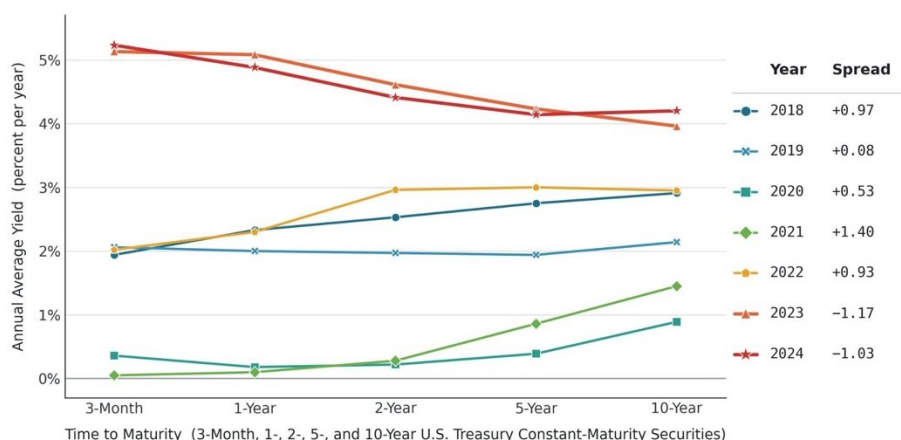
The post pandemic tightening cycle (2022-2024) provides a significant finding from the Taylor Rule analysis. With the average inflation rate at 5.02% and output gap of 2.40, the rule prescribed an interest rate of 9.71% — whereas the interest rate peaked at 5.33%. This created a policy gap of 4.38 points at the peak of the cycle. Despite the 525 basis points delivered in 16 months– the most aggressive tightening cycle in 40 years– the Fed policy remained below the rate prescribed by the Taylor rule through this regime.

There are assumption that underlie the prescription of these rates. Firstly the neutral real rate being fixed at 2% across the entire timeline of the study, even though the estimates suggested that neutral real rate was falling downwards during 2018-2024. A lower neutral real rate would reduce the prescribed rate for each regime. Secondly the prescribed rate for the zero-lower bound period exceeds the pre-pandemic figure (4.35%) only because the average CPI was at 2.97% which is mainly domination by the 2021 reflation. These oversimplifications warrant estimates of the Taylor rule and not precise prescriptions.

Figure 1: US Treasury Annual Average Yields by Maturity, 2018-2024 (FRED)

U.S. Treasury Annual Average Yields by Maturity, 2018-2024

Constant-maturity yields rose from a flat ~2% plateau, collapsed to near zero in 2020-2021, then inverted in 2023-2024



Source: Federal Reserve Economic Data (FRED). Values are annual averages of daily constant-maturity Treasury yields.

The effect of interest rates on JP Morgan's financial performance can be explained through three main channels: the net interest margin channel, the credit risk channel, interest rate risk channel

2.2.1) Net Interest Margin Channel

The net interest margin channel is a mechanism in monetary policy that explains how shifting interest rates influence the economy by affecting banks' profitability. For financial intermediaries like JP Morgan's profitability lies in the spread earned between yield on its assets and the cost of its funding. NIM is expressed as $(\text{Interest Income} - \text{Interest Expense Average}) / \text{Interest-Earning Assets}$.

It is also an essential component of PPNR relative to assets. The NIM channel functions as an asymmetric speed with response to a change in interest rates. Specifically when the Federal Reserve increases the benchmark rate, JP Morgan's assets reprice swiftly as floating rate commercial loans, credit-card balances and short term securities are indexed into a reference rate or mature into higher yielding assets in months. Whereas, the funding side reprices slower, as banks continue to delay raising deposit rates to expand their NIM. Besides that, a large share of JP Morgan's deposits around 15% of JP Morgan assets were non-interest bearing during the zero lower bound period. Even their interest bearing deposits also have a low deposit beta—the fraction of the rate hikes passed on to customers—consequently, NIM expands during a tightening cycle due to the constant increase in interest income while deposits costs lag. Apart from the beta below the asymmetric repricing strengthens the margin expansion: asset yield rising with the interest rates while the funding costs maintained a lag. On the contrary, this mechanism is reversed during an easing period as asset yields fall towards the floor and deposit costs have limited room to fall further. Specifically, the total deposits moved from \$2.622 trillion in 2018, to \$2.34 trillion in 2022 and finally in 2024 it was around \$2.4 trillion. Thus, explaining how the asymmetric mechanism shifts PPNR relative to assets during rate hikes/cuts.

Figure 1 illustrates this relationship by plotting the US Treasury term structure across the 2018-2024. In the first time frame (2018-2019) the yield is flat and plateaus at ~2% with both short and long term maturities offering similar yields. While in the second time frame (2020-2021) it reaches near zero levels indicating the policy rate collapse. However, by 2023 the curve is inverted because the short run assets provide a greater yield than the long run assets resulting in a negative spread.

2.2.2) Credit Risk Channel

The credit risk channel is a mechanism in monetary policy that explains how macroeconomic shocks (including interest rates), affect the economy by altering the perceived creditworthiness and the default risk of borrowers. Thereby, when the Federal Reserve increases its benchmark interest rate, the debt-service burden on borrowers significantly increases due to their obligation to pay higher interest on loans due to the variable repricing of their payments matching the higher policy rates. As these debt-service costs increase relative to a borrower's income the probability of default increases—in turn increasing the Net charge-off rate. Moreover, JP

Morgan's provisions follow the expected-loss standard adopted in 2020, resulting in an increase in the credit loss provisions in anticipation of deterioration of assets and higher rate of defaults.

2.2.3) Interest Rate Risk Channel

The interest rate risk channel is a mechanism through which fluctuation in the interest rate affect a bank's capital base, liquidity and profitability. Moreover, the channel operates on fixed rate assets.

When interest rate rise, the value of JP Morgan's fixed rate securities falls due to the newer bonds offering higher yields. In addition to that until the firm has not sold these securities resulting in a loss across its AFS and HTM portfolio which results in unrealised securities losses. Additionally this loss results in loss for two other indicators. Firstly capital: Given, JP Morgan's status as a G-SIB they cannot opt out of Accumulated Other Comprehensive Income (AOCI) in their regulatory capital. This results in the AFS losses flowing into CET1 lowering the capital buffer they have to sustain solvency. Even though, HTM losses do not directly reduce CET1 because the securities are held until they achieve maturity, yet they still represent a large economic loss. If liquidity strains forced the firm to sell its securities—meaning that the unrealized losses become realized and weaken the capital—this concealed exposure became evident in 2022Q3 for JP Morgan as their HTM latent losses reached \$40.3B, 19.2% of CET1. This highlights this risk of interest-rate risk that is entrenched in their balance sheet.

Secondly liquidity: because if funding reduces forcing JP Morgan to sell the securities to meet its obligations, the unrealised losses becomes a realised loss, reducing CET1. However, LCR is a more complex indicator as a direct rate hike does not result in an increase/reduction in the ratio. This is because of the diversified portfolio that JP Morgan has, even with the value of assets falling, the LCR can remain unchanged due to firm's liquidity management, funding structure, and depositor behaviour

Section 3

3.1) Spearman's Rank Correlation

Spearman Rank Correlation is used as a supporting statistical measure to quantify the relationship between the interest rate and the JP Morgan's financial performance. The coefficient measures the strength and the direction of the relationship of the dependent variable and independent variable on scale from -1 to +1. The results are summarized below in Table 2.

Table 2: Spearman Rank Correlation

Note*

- The Correlations are computed on the indicator before the direction coefficient (D_i) is applied. To ensure that a negative sign means an inverse relationship and a positive sign means a direct relationship.

- The reported p-values assume independent observations because the quarterly data may be autocorrelated, therefore the nominal significance shown here may overstate the true significance.

Indicator	ρ	p-value	Nominal Significance
Earnings – PPNR/Assets (+)	+0.601	0.0007	Yes (1%)
Capital - CET1 Ratio	+0.504	0.0062	Yes (1%)
Earnings + Capital Composite (+)	+0.722	<0.0001	Yes (1%)
Asset Quality -NCO Rate (-)	+0.513	0.0053	Yes (1%)
Liquidity - LCR (+)	+0.189	0.332	No
Unrealised Securities/CET1	+0.627	<0.0001	Yes (1%)
BFPI Without LCR	+0.111	0.575	NO
Full Five Indicator	+0.091	0.644	No

The two pillar BFPI (Earnings $\rho = +0.601$; capital $\rho = +0.504$) have a positive relationship of $\rho = +0.722$, a strong positive correlation, indicating that these indicators can indicate changes in financial performance for JP Morgan. By contrast, the four pillar BFPI excluding LCR is weakly correlated ($\rho = +0.111$) due to the downwards pull of securities and asset channels. Whereas, LCR is directionally consistent with the transmission channels discussed in Section II. Liquidity Coverage Ratio (LCR) shows no significant relationship ($\rho = +0.189$) illustrating the paper's claim that liquidity is mainly influenced by the funding structure of the firm and the deposit mechanism rather than interest rates directly.

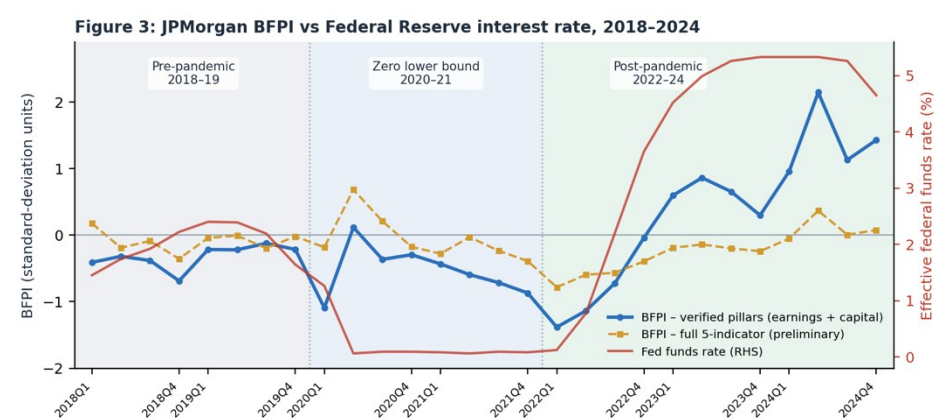
3.2) Regime Analysis

The tracking of the three composite regime substantiates that the relationship is sustained throughout a cycle rather than a set of outliers. Table 3 explains the average rate and two pillar BFPI robust z scores alongside the Taylor Rule Prescribed Rate from Section 2.

Table 3: Monetary Regime BFPI

Monetary Regime	Average Rate (%)	Earnings + Capital BFPI	Full BFPI	Taylor Rule i(%)
Pre Pandemic (2018-2019)	1.99	-0.32	-0.09	4.35
Zero Lower Bound (2020-2021)	0.23	-0.53	-0.05	4.71
Post Pandemic (2022-2024)	3.95	+0.40	-0.23	9.71

Figure 2: JP Morgan BFPI Vs Federal Reserve Interest Rate, 2018-2024



During the pre-pandemic regime the interest rate averaged around 2% consequently, the earnings-and-capital composite was significantly below its benchmark of -0.32. The earnings indicator averaged at around (+0.03) and the capital indicator at around (-0.67). While JP Morgan’s CET1 ratio was consistently around 12%, falling below its seven year median (2018-2024). This phenomenon can be explained by the net interest margin channel in Section 2.2.1. The Fed implemented a mild tightening cycle; hence, the realised rate was significantly lower than the normative benchmark set by the Taylor Rule, due to the Fed adapting policy relative to economic conditions and goals.

The zero lower bound period has the smallest mean out of all 3 regimes of -0.53 and the decomposition reverses during this period. Specifically, the earnings indicator as it falls to -0.91 while capital is near the neutral level of -0.15. This is due to the rate cuts resulting in new-loan yields repricing toward zero and the non-interest bearing assets cannot fall further. Moreover, as illustrated in Figure 2, the BFPI line has a lag behind the interest rate — overtime the assets begin to gradually reprice starting from the trough period of 2022, reflecting the asymmetric repricing lag of the margin channel.

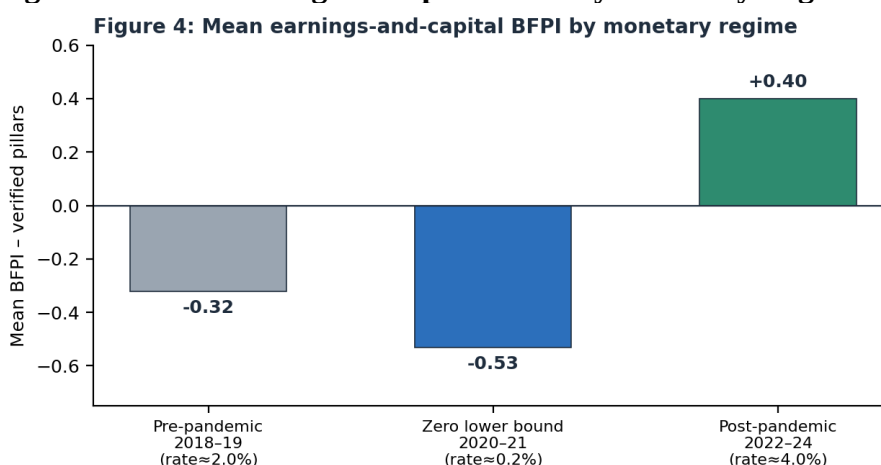
The post pandemic Period is strongest regime out of all 3 due to its mean of 0.40 — more than nine-tenths of a standard-deviation unit from the zero-lower-bound trough. Moreover, the initial expectation from this period is that the earnings indicator drives the financial performance because of the wider interest margin. Contrary to the expectation that rate hikes would make earnings the primary driver in the BFPI. However the modified z scores show that the earnings indicator was averaging at -0.03 (near zero levels); with indicators like capital averaging at +0.83, which caused the CET1 ratio to rise above 15%.

This is partially due to the acquisition of First Republic (1'st May 2023) added around \$200 Billion in assets and a vast risk weighted asset base, while lowering the CET1 ratio. This one time purchase produced \$2.6-2.7 billion after tax. Excluding the First Republic affected quarters, the regime means falls to -0.54 indicating the regime 's mean was elevated by the acquisition. Additionally, the financial performance is driven by the fast growing asset base and these quarters dominate others when benchmarked against the seven year median.

The margin channel explains the effect of the rate hike in this regime as seen in Figure 3 the monotonic climb began from the mean of -0.53 (in 2021) and rose until 0.40 (in 2024).

Although these regimes average's provide a comprehensive overview of the quarter by quarter effect of interest rates on financial performance; these averages must not be mistaken for plateaus. Given that, the post pandemic regime has the trough and the peak which compress the mean into more than 3.5 standardized units. As the index slides into near zero levels, gradually reprices and has sudden spike through the tightening cycles.

Figure 3: Mean Earnings & Capital BFPI By Monetary Regime

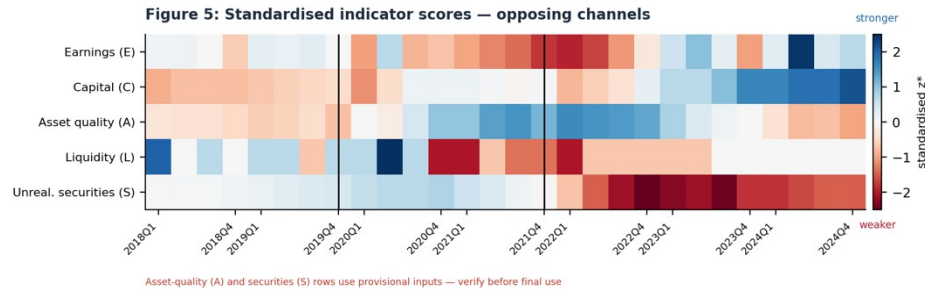


3.3) Composite Picture & Offsetting Channels

The full five indicator index provides a more sophisticated outlook, and it is presented as a preliminary finding because LCR are not directly taken from JP Morgan's investor reports and is sourced externally. The BFPI shows no significant association with the interest rate as ($\rho = +0.111$), indicating that the regimes do not follow the rate cycle — -0.09, -0.05 and -0.23 across three periods, with the high rate post the pandemic period recording the lowest composite score.

The central paradox arises when the two pillar core is compared against the full index. The earnings and the capital composite index has a strong positive correlation ($\rho = +0.722$), however the five indicator is uncorrelated ($\rho = +0.111$) with interest rates, reaching its lowest value of (-0.23) in the post pandemic regime where the rate were the highest. Therefore, it can be concluded that the influence of interest rates on financial performance is not uniform: It is positive relative with profitability and solvency but for asset value and unrealised securities it has an inverse effect, and interest rates have very limited effect on liquidity.

Figure 4: Heatmap of Standardised Indicator Scores

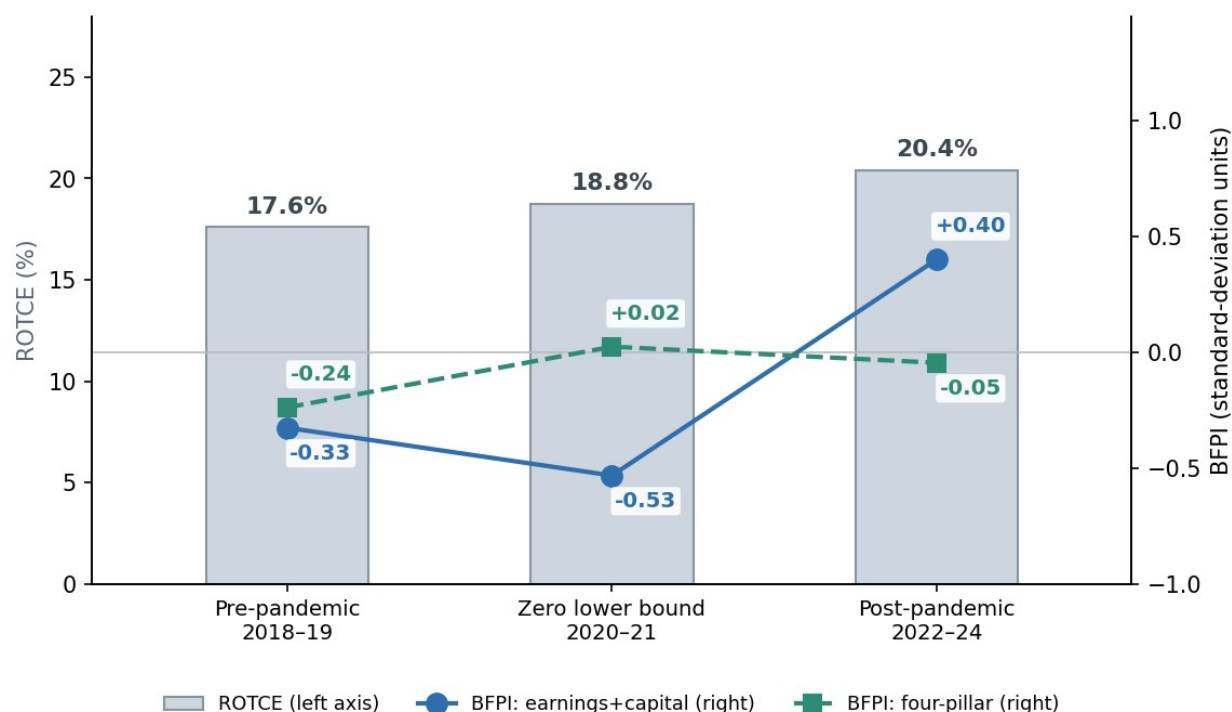


The heatmap above maps each channel onto its indicator. The earnings row starts off as a pale in the low rate years and gradually transitions into the strong blue during 2024 Q2. The capital row follows a similar path with a consistent lag as the retained earnings accrue into the CET1 ratio. Whereas the asset-quality row has a mild blue during zero lower bound period and cools toward red, as the net charge offs start normalizing during rate hikes. The liquidity row has no significant drift with the rate (except for 2020 Q4, 2021,Q1) matching its correlation. Finally, the unrealised securities row has the most evident shift as it was near neutral until 2021 and then eventually it transitioned into a strong red across the post pandemic regime. With rising yields marking down the fixed rate books.

Moreover, Figure 4 highlights how the five indicators move in opposite directions. During 2022-24 period as interest rates rose — the earnings and capital rows maintained a strong modified z score. Whereas the unrealised securities row shifted towards a dark red, due to the widening of the interest-risk-channel. The second quarter for 2024 illustrates this as earnings stood at +2.43 and capital was +1.86 z units. While the securities indicator was at -0.68 and the asset quality sat at -1.65. Consequently, the four pillar indicator average collapsed to just +0.49 z units even with the two pillar core reaching +2.15 z units. This margin expansion and the interest-rate-risk losses are not separate results, rather they are two consequences of the same rate hikes operating through two separate channels of Section 2.2

Figure 5: ROTCE VS BFPI Comparison By Monetary Regime

Figure 5: ROTCE and BFPI by monetary regime



To establish validity for the BFPI a comparison is conducted against Return on Tangible Common Equity (ROTCE). Across all 28 quarters, the four pillar BFPI has a correlation of $\rho = +0.65$ with ROTCE. This confirms that the index tracks profitability rather than random variations. Overall, the correlation is driven by the capital pillar with a correlation of $\rho = +0.62$ because retained earnings directly flows into CET1. However the pre-provision profit pillar only has a correlation of $\rho = +0.29$ because it excludes the credit provisions that dominate ROTCE especially during the zero-lower-bound period. Whereas, the securities pillar has a negative correlation of $\rho = -0.15$ because AFS losses directly accumulate into AOCI and evade net interest income. Therefore, it can be concluded that the index agrees with ROTCE on the profitability aspect. This is illustrated by figure 5 where ROTCE had a monotonic rise across all three regimes and the BFPI : Earnings + Capital pillar has a similar monotonic rise. However, there is divergence between the two in the zero-lower-bound period as ROTCE was at 18.8%; in contrast the BFPI pillar was at -0.53 z units. This divergence can be explained by the firm source of returns in 2021. As in this regime the ROTCE was significantly boosted by the one off credit releases that increased profits but were not a part of the banks operating performance.

Section 4 – Discussion

The findings situate JP Morgan within the literature that treats bank profitability with a positive correlation to the level of interest rate. Given the consistent rise of the two pillar core indicators ($\rho = +0.722$) is consistent with the framework given by Borio et al. (2017), who find that the

higher short term rates and steeper yield curve increase net interest income which results in an improvement in profitability levels. The main operative mechanism that is observed is the net-interest-margin channel of Section 2.2.1 and its strengths demonstrates the feature specific to JP Morgan: (Drechsler et al., 2017) provide an argument that in the deposits channel account, bank with a vast low cost deposit franchises can maintain low funding costs as policy rates rise. Thus, ensuring that asset yields can outrun the deposit costs and the margins widens. Moreover, JP Morgan's scale and extensive deposit base makes it amongst the premier beneficiaries that reflect the asymmetric nature of the yield curve.

The same rate hikes that widened the margin also operated through the interest-rate-risk channel. The 2023 regional banking stress demonstrates why these consequences can easily diverge across institutions. Silicon Valley Bank collapsed on March 10 2023 through the mechanism captured by the securities indicator. The concentrated portfolio filled with long term securities had lost its market value, as interest rates rose, as a result depositors withdrew their funds. This forced the bank to sell its holdings and transforming unrealised securities losses into realised ones, which exhausted the bank's capital. JP Morgan experienced a similar markdown on its AFS and HTM books visible on the balance sheet and the BFPI highlights how the securities indicator was negative for the post pandemic regime. Even though its capital was above the regulatory Basel III requirements and with the CET1 ratio well over 15% the losses reduced reported the capital without forcing a sale of the securities. This stark contrast isolates profitability and the capital composite indicators from the liquidity, securities and assets quality indicators. Therefore, it can be established that it was not the exposure to the interest-rate-risk channel but it was the size of JP Morgan's capital buffer. This allowed the firm to effectively to absorb its losses during the pandemic.

The Taylor-rule analysis in Section 2.2 effectively frames this pattern across all three time periods. Given that, the realised interest rate was well below the prescribed levels of the Taylor Rule—by 2.35 points pre-pandemic, 4.66 points during the zero lower bound and 4.38 points during the post-pandemic. This indicates the rate environment was systemic, yet it was constantly subdued.

Table 4: Comparison Against G-SIBs

Metric (FY 2023)	JP Morgan	Wells Fargo	Bank of America
HTM Amortized cost (\$ M)	369,848	262,708	594,555
HTM Fair Value (\$ M)	342,800	227,316	496,597
HTM Unrealized Losses(\$ M)	27,048	35,392	97,958

CET1	250,606	140,783	194,928
HTM/ CET1	10.8%	25.2%	50.3%
AOCI	10,433	11,580	17788
AOCI Loss/ CET1	4.2%	8.2%	9.1%
CET1 Ratio	15.0%	11.4%	11.8%
ROTCE	21.0%	13.1%	14.5%
Non-NII Share of Revenue	~44%	36.6%	42.2%
Non-Interest Bearing Deposit Share	26.8	~29	27.6

Table 4 compares JP Morgan against 2 two other large G-SIBs. The held to maturity losses relative to CET1 has the largest difference amongst all three banks. 10.8% for JP Morgan; 25.2% for Wells Fargo; 50.3% for Bank of America. This gap is a direct measure of the interest-rate-risk exposure, for every dollar in HTM losses that could weaken capital if funding shock forced a sale, this is the exact mechanism that resulted in a collapse for Silicon Valley Bank. Bank of America (BOA) had almost 5x HTM losses relative to CET1 of JP Morgan's reflects the BOA's decision to place a major share of their deposits in long-term held-to-maturity securities at low yields.

Table 4 helps explain why this risk is almost always overlooked. The AOCI loss reflected in the CET1 capital (4.2%, 8.2%, 9.1%) respectively for the firms. However, the greater concern lies is the HTM portfolio where the unrealized losses are substantially larger. While BOA's AOCI loss lies at ~9.1% on the flipside their HTM unrealised losses sit at 50.3% of CET1 indicating a greater level of interest rate risk exposure than what the CET1 numbers suggest. This comparison illustrates how focusing only on the capital can understate the overall balance sheet interest related risk.

On the other hand, the remaining indicators show that JP Morgan entered the post-pandemic regime (with interest rate hikes) with a greater position than their competitors. As it had the lowest securities exposure, the highest CET1 ratio (15.0%), the strongest profitability demonstrated by its ROTCE of 21.0%. Besides that, JP Morgan also boasted the most diversified revenue stream with 44% of its revenue stemming from Non NII assets & 26.8 % of its deposits were non-interest bearing.

These results highlight the differences between three G-SIBs JP Morgan allotted a smaller share of its assets to long term dated securities, all while maintaining a large capital buffer and ensuring that they have a low reliance on interest income. Consequently, these characteristics ensured that the firm could maintain low sensitivity while responding to interest rate changes.

Therefore, JP Morgan's experience and operating environment ensures that it is viewed as an upper bound classification of a G-SIB resilience.

Furthermore, the index here is used for one banking firm but the estimator is general, which ensures that the firm's median and median absolute deviation provides a self-calibrated specific to each firm comparing their performance against their previous year's performance. Therefore, framework is replicable for any firm comparing their own performance with themselves. However, a limitation arises that the performance of the firm cannot be standardized against its competitors.

Section 5 – Conclusion

This study's main purpose was to investigate the effect of interest rates on JP Morgan's financial performance between 2018 and 2024 using an original index, transmission channels and the spearman rank correlation to establish the relationship across three distinct monetary regimes. On one front, profitability and solvency are largely associated with interest rates contrastingly, for asset quality and securities there is almost no correlation between the variables.

Thereby two implications are created: Firstly, for policymakers, the gap between the policy rate and the normative rule based benchmark is a measure of latent interest-rate-risk within the banking system. For banks, the JP Morgan case provides an effective template to follow: a low cost deposit firm will widen its margins through tightening cycle, while the capital that is held above the regulatory requirements will convert the mark-to-market losses from threat into manageable costs.

However, these conclusions are bounded by the limitations of the study. The sole impact of interest rates cannot be isolated from the macroeconomic conditions that change with them; the modelling and median standardization choices for the BFPI and the use of limited data points for Spearman's rank correlation. Future literature can potentially quantify the deposits beta directly and make the BFPI usable to a vast panel of banks that differ in size and capital to develop a generic measurement index of financial performance.

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